



Green University of Bangladesh
Department of Computer Science and Engineering
(CSE)

Faculty of Sciences and Engineering
Semester: (Spring, Year: 2023), B.Sc. in CSE (Day)

LAB REPORT NO :04

Course Title: Artificial Intelligence Lab

Course Code: CSE-316

Section: 211D2

Lab Experiment Name: K-Means Clustering Algorithm

Student Details

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Lab Date : 31-05-24

Submission Date : 04-06-24

Course Teacher's Name : Md Naimul Pathan

Lab Report Status

Marks:

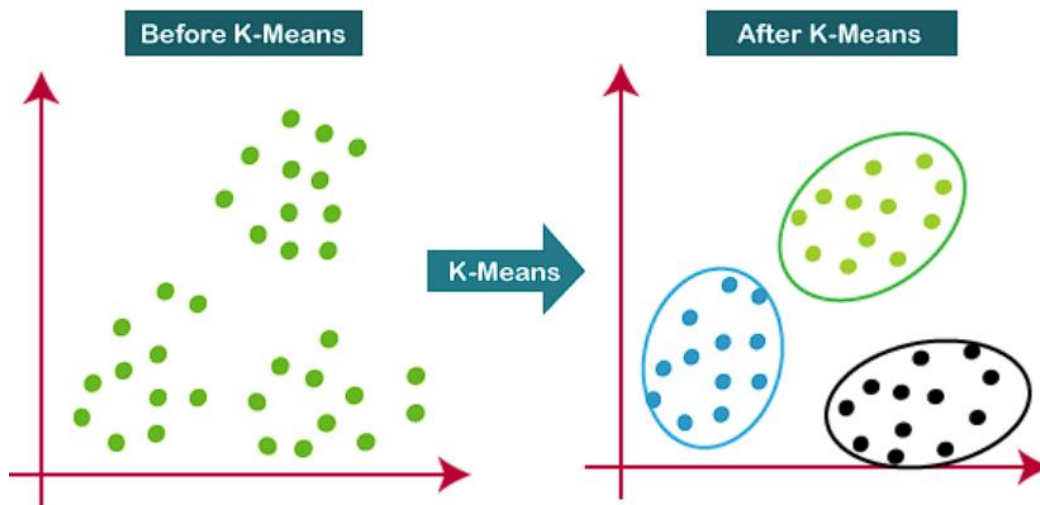
Comments:.....

Signature:.....

Date:.....

1. **TITLE:** K-Means Clustering Algorithm

2. **OBJECTIVES:** K-Means Clustering is an unsupervised learning algorithm that is used to solve the clustering problems in machine learning. The k-means clustering algorithm mainly performs two tasks:



The K-Means Algorithm Work:

The working of the K-Means algorithm is explained in the below steps:

Step-1: Select the number K to decide the number of clusters.

Step-2: Select random K points or centroids. (It can be other from the input dataset).

Step-3: Assign each data point to their closest centroid, which will form the predefined K clusters.

Step-4: Calculate the variance and place a new centroid of each cluster.

Step-5: Repeat the third steps, which means reassign each datapoint to the new closest centroid of each cluster.

Step-6: If any reassignment occurs, then go to step-4 else go to FINISH.

Step-7: The model is ready.

Elbow Method:

$$WCSS = \sum_{P_i \text{ in Cluster1}} \text{distance}(P_i C_1)^2 + \sum_{P_i \text{ in Cluster2}} \text{distance}(P_i C_2)^2 + \sum_{P_i \text{ in Cluster3}} \text{distance}(P_i C_3)^2$$

3. PROCEDURE:

Data Preparation	Perform any necessary data preprocessing steps such as handling missing values, feature scaling, and encoding categorical variables.
Used and alternative resources	Split both datasets into training and testing sets. Train various classification algorithms (K-Means Algorithms) on the training set of each dataset.
Unsupervised Learning (Clustering)	Apply the K-means clustering algorithm to both datasets. Visualize the clustering results using scatter plots, focusing on features such as Sepal Length vs Petal Width. Interpret the clustering results and assess their coherence and relevance.
Comparison and Analysis	Analyses any differences in model effectiveness, feature importance, clustering results, and scalability between the datasets.

4. IMPLEMENTATION: (Load The DataSet:)

<code>iris= datasets.load_iris()</code>
<code>x=pd.DataFrame(iris.data, columns=['SL','SW','PL','PW'])</code>
<code>y=pd.DataFrame(iris.target, columns=['Target'])</code>

Unsupervised Learning (Clustering):

<code>import matplotlib.patches as mpatche plt.figure(figsize=(12,3)) colors = np.array(['red','green','blue'])</code>
<code>iris_targets_legend = np.array(iris.target_names) red_patch = mpatches.Patch(color='red',label='Setosa')</code>
<code>green_patch = mpatches.Patch(color='green',label='Versicolor') blue_patch =</code>
<code>mpatches.Patch(color='blue',label='Verginica')</code>

Iris k-mean model cluster Model:

<code>iris_k_mean_model = KMeans(n_clusters=3,n_init=10) iris_k_mean_model.fit(x)</code>
<code>print(iris_k_mean_model.predict(x)) print(iris_k_mean_model.cluster_centers_)</code>

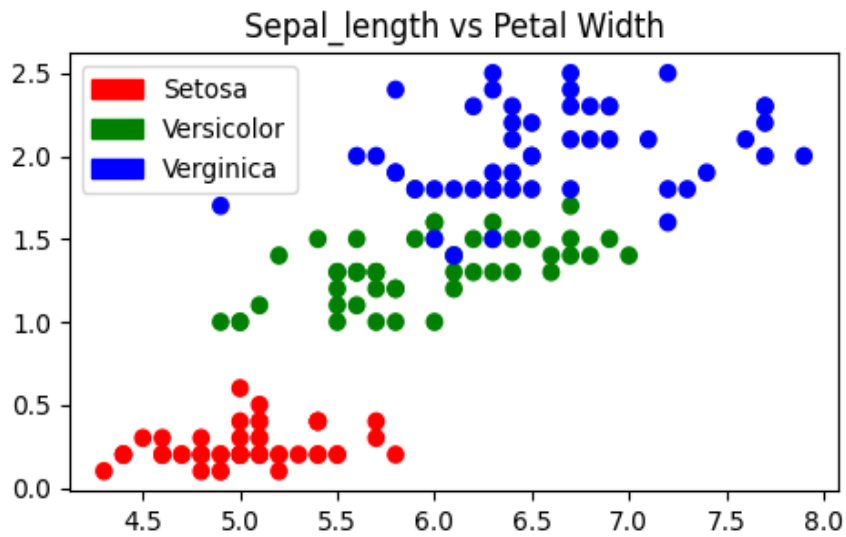
Find accuracy of the trained model:

<code>plt.subplot(1,2,2) plt.scatter(x['SL'], x['PW'], c=colors[predictedY]) plt.title("Models classification")</code>
<code>plt.legend(handles=[red_patch, green_patch, blue_patch]) print("Accuracy Score:"+</code>
<code>str(ss.accuracy_score(predictedY,y['Target'])))</code>

5. RESULT:

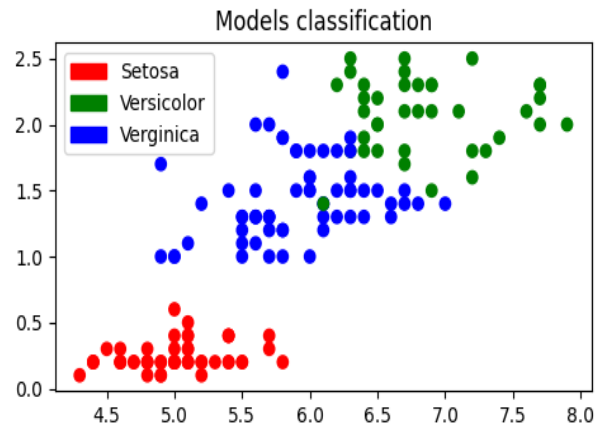
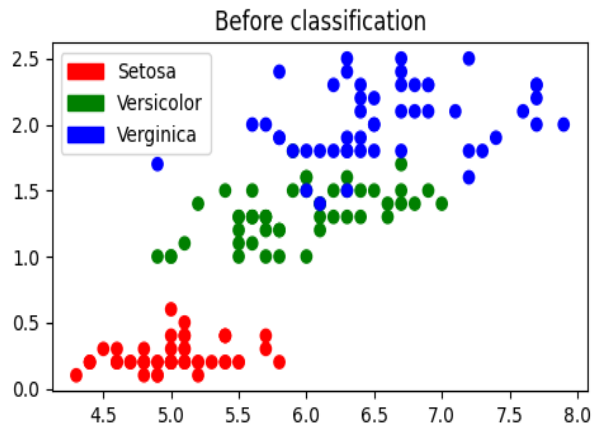
Iris k-mean model cluster Model

Visualize the data using scatter plots:

[illegible]

Check the classification performance of the trained model

Check the Accuracy and Array:



Accuracy Score:0.44

```
array([[50, 0, 0],  
       [ 0, 2, 36],  
       [ 0, 48, 14]])
```

6. ANALYSIS AND DISCUSSION: In conclusion, K-means clustering is a powerful unsupervised machine learning algorithm for grouping unlabeled datasets. Its objective is to divide data into clusters, making similar data points part of the same group. The algorithm initializes cluster centroids and iteratively assigns data points to the nearest centroid, updating centroids based on the mean of points in each cluster.